



Review

Just transition: A conceptual review

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ABSTRACT

The growing attention paid to the idea of a just transition away from the incumbent fossil fuel energy paradigm has led scholars to devise diverse definitions, understandings, and viewpoints of the term. This review seeks to clarify the different perspectives surrounding the concept, to consolidate knowledge, and to provide a concise account of current debates in the literature as well as a research agenda. It identifies five themes around which the concept has been discussed: (1) just transition as a labor-oriented concept, (2) just transition as an integrated framework for justice, (3) just transition as a theory of socio-technical transition, (4) just transition as a governance strategy, and (5) just transition as public perception. Overall, this review suggests that the literature on just transition employs rich theoretical and empirical insights from various disciplines yet contains several gaps. Specifically, it argues that the literature would benefit from more empirical studies rooted in practice, more discussion on the relationship between different concepts of just transition, an expansion of geographical scope to include developing countries and non-democratic regimes, and more attention to power dynamics in just transition.

1. Introduction

Addressing the threat of anthropogenic climate change requires a fundamental transition from the incumbent fossil fuel energy system to a low/zero carbon system [1–3]. Rather than completely shifting from one set of dominant energy resources to another, energy transition involves the phasing out of fossil fuels, while increasing renewable energy resources such as solar and wind, which used to occupy only a small share of the energy mix [4]. This shift necessitates complicated social and technological changes and will have a profound social impact. Scholars have indicated that such a shift must consciously avoid reproducing the environmental and socio-economic injustices and inequalities which are intrinsic to the existing energy regime [4–8]. Indeed, the low-carbon transition is a multifaceted phenomenon entailing considerable benefits and risks [9]. The benefits of shifting to a less carbon-intensive socio-technical regime include more resilient and democratic energy systems [10], industrial restructuring and technological advancements [11], poverty alleviation [12], and addressing social and environmental justice concerns within and across nations [13]. The development of renewable energy is considered a crucial tool for advancing sustainable development goals, as it could replace the highly polluted, dangerous, grey economy with an environmentally friendly, healthy, green economy [13].

The costs and risks of such a fundamental transition, however, are troubling. The decline of carbon-intensive industries has caused difficulties for individuals who held fossil fuel jobs. They often encounter difficulties in finding a new job because of a skills gap or are forced to make sacrifices such as substantial wage loss and long-distance commutes, along with compromises in culture, community identity, and sense of place [4,14]. Moreover, fossil fuel industries play an important role in the local tax revenue base; the decline of the industry may adversely affect public services such as education, transportation, and waste management [4]. Furthermore, the social consequences of labor disruptions extend to other employment sectors within fossil fuel-based communities [4].

Justice concerns in the low-carbon transitions arise not only from the fossil fuel industry decline, but also from the inequitable distribution of benefits. Studies indicate that renewable energy, the so-called “green jobs creators,” may hardly benefit marginalized groups. The representation of women and people of color in the United States solar industry workforce is well below the national average for all occupations [4]. In Germany, the development of solar energy shares many of the same inequalities that characterize fossil fuel industry, resulting in the creation of localized “sacrifice zones” [15]. On a global scale, the trend of male dominance in highly technical, well-paid jobs remains a constant in the renewable energy sector [4]. Strikingly, the disproportion

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distribution of low-carbon energy technologies and rising electricity prices caused by such transitions raises significant concerns for environmental justice [4,16] and energy poverty issues [17–19].

Therefore, the low-carbon transition is certainly not a magic elixir for a more “just” world despite its environmental and economic benefits [20]. As Sovacool et al. [9] rightly point out, the current wave of low-carbon transitions may “create new injustices and vulnerabilities, while also failing to address pre-existing structural drivers of injustice in energy markets and the wider socio-economy.” It is therefore imperative to shift the technology-focused understanding of low-carbon transition to include greater engagement with social justice issues [21]. As Williams and Doyon argue, “we cannot achieve a sustainability transition without justice, indeed that an unjust transition is not sustainable” [22]. In this context, the concept of just transition, which draws attention to the equity and justice issues associated with efforts to address energy and climate problems, is increasingly recognized as an important component of low-carbon transition regimes [23–27].

The concept of a just transition has a long history and has been applied by activists, labor unions, and associated groups dating back to the early 1970s [23,25], while scholarly debates on transition appeared more recently in the 2000s [28]. Yet, in a relatively short time, this concept has spawned an expansive body of literature, and scholars have committed significant efforts to the conceptual development of just transition [8,24,29]. In particular, multidisciplinary engagement with the concept has led to divergent interpretations, resulting in multiple and ambiguous definitions [27,30,31]. There is no universally agreed-upon definition of or framework for just transitions [31]. The concept of just transition has become, in our view, so ambiguous and with so many different meanings that communication and debate have become difficult. An overview and consolidation of academic literature around just transition is urgently needed to clarify these different conceptions and their inter-relationships. Another motivation of this study is to identify the limitations of existing research in just transition and to suggest a research agenda.

A few reviews engage with the concept of just transition, but they are often limited to one particular strand of the literature. For instance, in their conceptual review, García-García et al. [32] review the just transition literature with specific attention on the impact of the low-carbon transition on labor income. Jenkins et al. [33] focus on the applications of the energy justice framework in low-carbon transition studies. This approach is echoed by Sovacool and Dworkin [34], who develop energy justice as a conceptual tool, analytical tool, and decision-making tool for three groups of people: philosophers and ethicists; energy researchers; energy planners and consumers. From a climate justice perspective, Sonja and Hadi [35] provide a conceptual review with a specific focus on distributive justice implementation and its potential outcomes in international climate policies. Because the existing reviews mainly focus on one interpretation, they do not reflect the diversities of just transition and the relationship among the different dimensions of just transition.

This paper provides a comprehensive conceptual review of the just transition literature. The method of review is based on the systematic review approach typically adopted in the field of energy studies [32]. First, to determine the objective of the study, an analysis of the most referenced articles, particularly review papers, was conducted. This led to the realization that a review of just transitions to clarify the different interpretations was needed. Second, we employed an inclusive search strategy to retrieve papers containing the different conceptual interpretations. Keywords adopted were not restricted to “just transition,” but also related concepts including “justice,” “equity,” “fairness,” and “ethics” in conjunction with “energy / low-carbon / clean / sustainable / green transition.” These keywords were used to search databases including Web of Science and Google Scholar. We then searched additional references by checking the bibliography of the papers. Third, the publications collected were checked to ensure their relevance and analyzed through an inductive approach to thematic analysis. Then, a qualitative synthesis of the themes was conducted. Our search was not

limited by geographical locations or disciplinary boundaries. Geographically, most studies were carried out in North America, followed by China and Europe. In terms of the place of publication, the top contributor is *Energy Research & Social Science*, followed by *Energy Policy* and *Applied Energy*. Table 1 summarizes the reviewed materials included in our review.

The rest of the paper is structured as follow. From Section 2–6, we discuss each theme of just transition in detail. In Section 7, we reflect on the insights gained from the review exercise and recommend some directions for future research. Section 8 concludes the paper.

2. Just transition as a labor-oriented concept

In the first body of work in just transition, scholars adopt a historical approach to the development of the concept and further examine how the concept can be applied in the contemporary context. Many scholars have highlighted how the just transition concept emerged from labor movements and thus sees energy transition and economic transition as intrinsically linked [4,23–25,27,29,31]. The background is the strengthening of environmental governance in the 1970s negatively impacted employment in the industries that struggled to adapt due to their failure to meet environmental standards [23,25]. Labor unions have long played an important role in advocating for workers who face mass unemployment. Rather than fight for the repeal of environmental protection laws, the Oil, Chemical, and Atomic Workers Union (OCAW) saw the fate of the environment and of the workers as intertwined and decided to advocate for a just transition [24,32]. In the face of widespread unemployment, the OCAW proposed developing a “superfund” for worker retraining and community support [25,31]. Workers saw the appeal of just transition as a labor-oriented concept that could ensure that the need for environmental protection did not prevail over the well-being of workers and communities [26].

The original transition strategy emphasized the development of social protection programs for workers and communities in hazardous industries. This concept broadened during the 1990s as the OCAW allied itself with environmental justice groups across the United States [25,31]. However, the concept of a just transition did not gain traction in the United States because of the anti-environmentalist political culture [25]. Rather than allow it to disappear, activists and labor unions extended the concept’s scope [25]. They diffused just transition principles internationally by participating in global environmental negotiations starting in the early 1990s [24]. Since then, labor unions have been increasingly regarded as important stakeholders in global

Table 1
Top-15 ranking of journal, geographic focus, and year of publication.

Rank	Journal (number of publications)	Geographic focus (number of publications)
1	Energy Research & Social Science (22)	United States (13)
2	Energy Policy (18)	Europe (13)
3	Applied Energy (7)	China (7)
4	Research Policy (3)	Australia (3)
5	The Extractive Industries and Society (3)	Japan (2)
6	Environmental Politics (3)	South Africa (2)
7	Environmental Justice (2)	The Global South (2)
8	Environmental Innovation and Societal Transitions (2)	Mexico (1)
9	Geoforum (2)	South Korea (1)
10	Journal of Environmental Policy & Planning (2)	Nepal (1)
11	Sustainability (2)	India (1)
12	Climate Policy (2)	Indonesia (1)
13	Journal of Environmental Planning and Management (2)	Mozambique (1)
14	Wiley Interdisciplinary Reviews: Climate Change (2)	Pakistan (1)
15	International Environmental Agreements: Politics, Law and Economics (2)	The Developing World (1)

environmental negotiations as sustainable development discourse and policies have grown in prominence and have largely advocated for a just transition that includes “decent green jobs” [24]. This mantle has been taken up by climate justice activists in recent decades [25,31]. Together, these movements contributed to the acknowledgement of a need for a just transition within the final agreement of the 2010 United Nations Climate Change Conference [24]. Today, the scope of just transition discourse encompasses a range of issues, including food justice, green gentrification, and energy sovereignty [31].

The concept of a labor-oriented just transition is increasingly intertwined with calls for fossil fuel-dependent countries to develop future-oriented climate policies and economic development plans [23,25,31]. These calls seek to garner broad public support, break the fossil fuel industry’s stranglehold on energy policies, accelerate the pace of a transition away from an extractive economy to a regenerative one, and ultimately mitigate the severe economic hardships that fossil fuel-dependent communities and workers will face in the future [36,37]. Policies encouraging the growth of renewable energy have become a predominant remediation means for workers and communities affected by declining fossil fuel industries. However, the development of renewable energy may be contested in regions where the development are meant to aid, as fossil fuel-based communities and workers may feel threatened by the societal shift toward clean energy [36]. Furthermore, the definition of a just transition is often contested in practice, especially in the context of multi-actor policy networks. While labor unions advocate for finding or creating decent, well-paying jobs that are equivalent to the jobs that fossil fuel workers and others in their communities have lost, environmental groups emphasize the need for sustainable ecological outcomes [27]. However, a study exploring the transition away from coal production in Australia’s Latrobe Valley—a heartland region that generates 85–90% of the country’s electricity—following the shutdown of the Hazelwood power station found that, despite inconsistent definitions of what a just transition entails, invoking the concept “has served to open up a meaningful discussion between unions, environmentalists, and others on how best to mitigate the social fallout from necessary environmental policy changes and generate jobs” [27].

Advocates for a just transition do not see a conflict between environmental protection and economic vitality because they suggest that more jobs could be created in an environmentally friendly economy [24]. These advocates aim to foster an alternative socio-environmental politics which values collective representation by workers and communities around principles of social and environmental justice [28]. More recently, Henry et al. suggest that the ongoing COVID-19 pandemic might represent a catalyst for community and economic recovery efforts that support “a shift to a low-carbon energy system that prioritizes sustainability and social and economic equity” [31]. The optimistic attitude is also echoed by White, who points out that as the understanding of the labor-based concept increasingly has been expanded by labor, socialist, climate justice, environmental justice, feminist, and decolonial factions, this expansion could contribute to enrich the meaning of justice and to facilitate just transition alliances in low-carbon transitions [38]. Based on this observation, White proposes using the concept of just transition not only to enhance coalition-building capacity and to mobilize greater engagement of a range of social justice advocacy organizations, but also as a useful tool for public dialogues, allowing a very different kind of technical, cultural, political and economic knowledge to be integrated into the transition discursive formations [38]. However, opposing this optimistic view, Kalt argues that just transition as a worker-centered concept would lead to a slowdown of low-carbon transition [39]. In examining the historical impact of labor and climate movements on the jobs versus climate conflicts, Kalt points out that just transition has become a compromising outcome as climate movements failed in attracting labor union support, and labor unions remain allied with the fossil fuel industry to protect high-paying industrial jobs [39].

Studies adopting the labor-oriented approach typically focus on the agency of national and global labor unions [28]. However, while labor unions have indeed played a vital role in advocating for a just transition, they should not be perceived as a homogenous entity; rather, there are apparent tensions within and between trade unions about which and whose jobs they should be protecting and at what cost. Furthermore, business groups often mobilize labor unions to oppose ambitious climate action or support carbon intensive infrastructures (such as airports), and the role of government as a powerful actor regulating the relationship between businesses and labor unions cannot be understated. Therefore, it is also important to explore under what conditions and by what strategies these stakeholders engage in the politics of just transition and with what results.

3. Just transition as an integrated framework for justice

In the second strand of the literature, scholars attempt to transcend the historical meaning of a just transition described above by recognizing the linkage between the concept with other, more established concepts in the justice-related literature, including environmental, climate, and energy justice. We discuss each of the three concepts in turn to highlight these integrative attempts.

3.1. Environmental justice

The concept of environmental justice is historically grounded in the environmental justice movement, which has a long history of being at the forefront of fighting for vulnerable groups that are disproportionately impacted by environmental degradation [40–42]. Advocates for environmental justice argue that disadvantaged communities should not shoulder a disproportionate burden when it comes to the damage caused by environmental issues [22], and all citizens should be equally involved in the development, implementation, and enforcement of environmentally just policies [43]. Schlosberg and Collins note that the environmental justice movement has a broad agenda that links environmental and other social justice concerns such as the occupational health and safety movement, the indigenous land rights movement, the public health movement, and various social and economic justice movements, through building alliances with labor unions, environmentalists, and civil rights groups [19]. Additionally, environmental justice activists bring a wide variety of experiences, that in turn enrich the theoretical development of justice [19,44–46].

With the recognition that the causes of environmental pollution are rooted in political, economic, and cultural inequalities, the environmental justice concept takes a more holistic view to identify who is affected, and in what ways and how to address those inequities [14,40]. Consequently, in addition to its initial focus on the inequitable distribution of environmental risks, the elements of recognition and participation have been integrated into environmental justice advocacy as crucial demands [19,42]. Scholars highlight that recognition justice is an inherent precondition for distribution justice because without knowing who is affected, and in what ways, a fair distribution of benefits and harms cannot happen [42]. As Fraser [47] claims, “only by looking to alternative conceptions of redistribution and recognition can we meet the requirements of justice for all.” In addition, recognition is an integral part of procedural justice because no one can participate without being recognized and respected [42]. In turn, democratic and participatory decision-making procedures could provide solutions to address inequitable distribution and ensure substantive participation for affected vulnerable groups [42]. Overall, distribution justice, recognition justice and procedural justice are seen as interlinked and inseparable [42,48,49].

Given that the environmental justice framework has been expanding topically and geographically over decades, it has a large influence on the development of other justice-related concepts, including climate justice [19,50] and energy justice [21,51,52]. For instance, Schlosberg and

Collins elaborate on how a concern with the impact of climate change has been present in the environmental justice movement through an analysis of the relationship among environment, climate, and justice [19]. Regarding the close connections between environmental justice and energy justice studies, scholars [51,53] recognize that the two research areas not only share the three common justice principles (distributional, recognition and procedural justice), but also regard energy justice as part of the evolution of environmental justice [21]. However, unlike the concepts of environmental justice, climate justice, and just transition, the activism roots of energy justice are not always adequately recognized by scholars [54], leading Galvin to inquire “whether there is now a place within energy justice literature for studies based on the civil rights type of understanding of justice” [55].

It is also evident that the concept of environmental justice has become closely related to just transition, both theoretically and practically [56]. Historically, the environmental justice concept has been criticized as focusing too much on defending against environmental threats, whereas less emphasis has been placed on socioeconomic issues like job security and economic equity. This leads to a lack of capacity to engage with fossil fuel dependent communities, workers and their unions, thereby weakening the ability to unsettle the hegemony of the fossil fuel industries [57]. These challenges have led Evans and Phelan to argue that the environmental justice communities “simply advocate shifting hazardous industries from one locality to another,” and fails to ease the perceived tension between economic vitality and environmental protection as a result. Recently, scholars and practitioners of environmental justice have expanded their work to not merely responding to immediate environmental hazards, but to include future-oriented approaches to socio-economic justice and sustainability, such as shifting from an extractive to a regenerative economy. For example, the Just Transition Alliance in the United States brought together numerous environmental justice organizations to conceptualize and frame a more integrated approach to just transitions. Likewise, local organizations in coal-producing regions such as Wyoming and New Mexico emphasize labor-focused just transition frameworks often use environmental justice principles in their advocacy, such as a focus on “more participatory, deliberative, non-hierarchical sets of relations between states, citizens and communities” [58]. As such, both environmental justice and just transition frameworks now encompass, but also extend beyond the concerns of fossil fuel producing communities, such as including people who have been living with higher levels of pollution because of the disproportionate location of fossil fuel facilities in marginalized communities [14]. This conceptual development suggests that weaving together environmental justice and just transition could attract the support of those who may be vulnerable and marginalized in a post-carbon economy and pose a powerful and united front to the fossil fuel bloc [57].

3.2. Climate justice

The concept of climate justice has evolved in step with the concept of environmental justice, as they both share activist roots [22]. The prominence of the climate justice movement is reflected in the fact that it brings a large and diverse array of groups together that seek to challenge the systems that create climate change and impose disproportionate environmental costs and risks on less powerful groups [59]. Harris claims that climate change is “a matter of justice” [60], and he defines climate justice as a broader understanding of justice in terms of the “fairness, equity and rightness” of responses toward climate change issues [60]. Klinsky et al. summarize the four reasons that justice and equity are centered in climate change:

Obligation to address human wellbeing...; Understandings of justice are essential to political analysis...; Equity is not always in tension with strong climate action or collective action...; Understanding trade-offs requires taking equity into account [61].

Climate justice is used routinely to frame discussions about historical responsibility, finance, loss and damage, human rights, and adaptation [62–65]. Advocates for climate justice are not only concerned with environmental issues but also acknowledge that climate crises exacerbate existing social inequalities and, therefore, advocate for policies that mitigate the effects of climate change and adapt society to the emergent reality of climate change over the long term. Okereke highlights the three key elements of climate justice as follows:

Compensation for past harm (compensatory justice), distribution of future and present responsibilities (distributive justice), and adoption of fair procedures and inclusive framework in the process of reaching decision making (procedural justice) [66].

Many of these advocates perceive climate justice in the context of the uneven distribution of global warming’s effects on developing countries. They argue that countries that “contributed the most to the occurrence of the situation are more prepared” to deal with its effects than those that did not [26], and that impoverished people in developing nations who are least responsible for the climate crisis will suffer more from its effects than individuals in wealthy nations who lead high-carbon lifestyles [28]. They also argue that the people who will be most affected by climate change participate the least in the global decision-making process regarding climate change, which raises issues of procedural injustice [26]. Moreover, intergenerational justice is one of the key ethical concerns of climate justice because the future generation will be the most affected by the climate catastrophe but excluded from the relevant policy-making process [66].

Various climate justice organizations, such as the Climate Justice Alliance, Climate Justice Now, and Movement Generation, have contributed to the integration between climate justice and just transition. However, many countries are reluctant to take positive action to limit global warming. In this context, Routledge et al. (2018) point out that climate justice activists could learn lessons from just transition movements insofar as the latter’s emphasis on grassroots organizing can cultivate “more participatory, deliberative, non-hierarchical sets of relations between states, citizens and communities” and open up possibilities for just and sustainable futures [58].

3.3. Energy justice

Tackling climate change requires drastic changes in energy production and consumption. However, the distribution of harm and benefits within the incumbent energy system are unequal; thus, there is growing recognition in the literature that a just transition to a low-carbon future requires just trade-offs between “energy and development and carbon growth” so that all nations can benefit equally from the transition [23]. Theorizing these trade-offs is the domain of energy justice—the pursuit of “a global energy system that fairly distributes both the benefits and burdens of energy services, and one that contributes to more representative and inclusive energy decision-making” [67]. Energy justice scholars attempt to develop a framework for energy justice that can be applied in practice [63]. Such a framework evaluates “where injustices emerge, who is affected or ignored, and what processes exist for remediation to reveal and reduce injustices” [22]. Such frameworks are undergirded by four tenets: procedural justice, distributional justice, justice as recognition, and restorative justice [68]. The energy justice literature covers various energy-related issues, including energy poverty, large-scale energy projects, and energy transition [54,69–71].

Some scholars criticize the three-tenet (distributional, procedural, and recognition justice) approach to energy justice. For example, Lee and Byrne point out that the energy justice framework contains limitations on understanding the complicated social justice issues [21]. They argue that because the driving forces of injustice caused by the economic and political structures are largely missing in energy justice research, the function of the energy justice framework more likely focuses on

identifying injustices, rather than solving them [21]. Thus, Lee and Byrne add structural and ideological components, namely industrialism, technicism, and capitalism, to analyze why particularly vulnerable groups are always negatively affected by large-scale energy development models, and in what way the top-down, authoritarian decision-making approach serves as a political tool to support the implementation of these large-scale energy projects without addressing justice issues in the policymaking process [21].

In addition, energy justice scholars have started to rethink the problems created by a framework dominated by Western justice theorists. Sovacool et al. note that the justice philosophical underpinning of energy justice scholarship “relied almost exclusively on Western philosophers such as Jeremy Bentham, Immanuel Kant, John Rawls, and Robert Nozick” [67], and this led them to propose a new research agenda for future energy justice research that brings in non-Western philosophy of justice theories, such as Confucianism and Taoism, Hinduism, Buddhism, and indigenous perspectives of the Americas, to advance energy justice principle [67,72]. Furthermore, studying energy transition issues in Mozambique, Castán Broto et al. find the problem in applying energy justice principles to non-Western countries is that western tradition-based energy justice principles lack a contextual understanding of socio-spatial factors that shape local people’s energy needs [73]. Thus, they call for integrating the energy sovereignty notion into energy-related research, which refers to the core emphasis that “the right of people to access energy resources within ecological limits to have a dignified life.” The last important reflection on energy justice research also comes from Sovacool et al. [67], who state that energy justice should move forward from a human-centered approach to non-human-centered and from territorial-bounded, energy-focused approach to a global and intersectional approach [67].

Some scholars argue that the energy justice frameworks would benefit from a stronger engagement with the just transition concept [22]. For instance, Healy and Barry point out that “some energy justice literature underemphasizes the political economy of socio-technical energy transitions, particularly surrounding extraction” [7]. They claim that by integrating the concept of just transition into energy justice approaches, scholars might theorize social benefits and changes beyond the energy sector.

3.4. Just transition as an integrated framework

Each of these three justice frameworks (environment, climate, and energy) emphasize distributive justice and procedural justice and agree that despite the urgent need for a transition, such a transition must be just and equal in its effects and implementation [56]. However, it has been argued that because each of these frameworks treats their concepts as separate constructs, a joint conceptual space for reflection is absent [56]. Therefore, scholars propose developing a new analytical framework, which unites these three perspectives with the labor-oriented justice concept. Different approaches have been taken. For example, some scholars suggest creating a multidisciplinary justice-based framework [6], and others propose combining all three perspectives on justice under a just transition framework [43,56]. Still others suggest a more practical focus from the perspective of legal geography; one that emphasizes taking legal action to reduce inequality and injustice during the period of energy transition [2]. Williams and Doyon suggest that approaches to transition, environmental justice, and energy justice have not adequately explored recognition-based justice, and thus produce an analytical framework for assessing “issues of power” in this domain [22].

The concept of just transition thus offers an open space for developing an interdisciplinary approach to generate justice-oriented social formations in the context of a changing climate [56]. Combining the fundamental principles of justice—distributional justice, procedural justice, justice as recognition, and restorative justice—is key to developing a robust framework through which to analyze the possibilities for

a just transition. However, these frameworks are primarily normative and theoretical, and there is a lack of research that explores the practicality and challenges of applying the proposed frameworks in real situations.

4. Just transition as a theory of socio-technical transition

Socio-technical transitions refer to deep structural changes in systems that involve long-term and complex reconfigurations of technologies, policy, infrastructure, scientific knowledge, and social and cultural practices [23]. Theories of socio-technical transition seek to understand the trajectory of technological change and to reveal the origins, patterns, and mechanisms that drive them [22]. The concept of socio-technical transition emerged in literature in the 1990s. It suggests that technological transitions are more likely to occur alongside radical shifts in society’s structure, values, and use of technology; thus, the socio-technical transition framework explicitly engage with the complexity and uncertainty of transitions and that they might lead to fundamental changes in life and society [74].

The multi-level perspective is an influential analytical framework for understanding the multi-dimensional complexity of socio-technical transitions [75]. The multi-level perspective conceptualizes transitions as dynamic processes driven by three analytical levels: sociotechnical regimes, niche-innovations, and sociotechnical landscape [76,77]. The three analytical levels are summarized by Geels et al. as the socio-technical system (i.e., the dominant existing system), niche innovations, which differ radically from the existing system, and exogenous developments that destabilize the system and facilitate the breakthrough of niche innovations [77].

Geels et al. also point out that the multi-level perspective does not pay sufficient attention to the distribution of social benefits and burdens or to social sustainability issues such as inequality, poverty, and work conditions [78]. The approach’s focus on the dynamic processes of transitions means that social outcomes or impacts of such transitions are ignored, as transition scholars typically take it for granted that “green innovations are intrinsically positive” [78].

Transition management is another commonly adopted approach to study socio-technical transitions. The framework is a reflexive and participatory governance concept that seeks to manage changes in the desired direction by combining long-term thinking with short-term action through a way of learning by doing and doing by learning [74,79]. There are many advantages to adopting this approach including keeping track of the transition process to avoid undesirable deviations and to ensure the coalition and engagement of relevant stakeholders [33]. However, some scholars assert that this specific approach is difficult to apply in practice and the effectiveness of this approach cannot be assessed [33].

These analytical frameworks have been applied to a variety of issues in the energy transition literature, such as sustainable development and low-carbon energy transitions [76,80–86]. However, these studies have been criticized for their narrow focus on transition management, which is inconsistent with the disruptive and systemic-structural nature of socio-technical transformations [7]. Because they entail systemic change, energy transitions inevitably bring the danger of unequal distribution of benefits and losses. Thus, some scholars have suggested that considering this unequal distribution is a key component of “ensuring that system transitions are not only more sustainable, but also more just” [22]. Because the transition literature does not necessarily address “equity, justice and the political economy of transitions,” scholars have called for increased focus on these efforts [5,22].

These calls have led to the emergence of “just energy transition,” a discourse that centers social justice in energy transition to respond to the dual challenges of climate change and sustainability [87]. For example, García-García et al. view just energy transition as a renewed vision of just transition [32]. They define a just energy transition towards a low-carbon economy as:

...a long-term technological and socio-economic process of structural shift that affects the generation, distribution, storage and use of energy and causes rearrangements at micro (innovation), meso (social networks, rules and technical elements) and macro (exogenous environment) levels, while also ensuring that the desired socio-economic functions can be accomplished through decarbonised and renewable means of energy production and consumption, safeguarding social justice, equity and welfare [32].

Thus, the global low-carbon transition is such an extraordinarily tangled and complicated phenomenon that not only the sociotechnical matters—the profound structural change requires technology and innovations, while social justice aspects are also critically at stake in such transformations [49]. While the two dimensions, socio-technical transition and social justice implications, have been treated as distinct scholarship in the past, some scholars have recently attempted to bridge this gap. For example, Jenkins et al. [88] propose a hegemonic framework to combine justice principles with socio-technical transition studies. By arguing that justice concerns would be difficult to fully consider among other complex social contexts through the multi-level perspective framework, Sareen and Haarstad [49] propose an inclusive approach to synthesize the strengths of social justice research and social-technical literature, instead of developing a new framework by combining energy justice with socio-technical transition theories.

Other scholars have addressed the question of how to achieve a just energy transition in recent years. Heffron et al. highlight that industrial demand could help attain a just energy transition given industry's flexible potential “to address short-run and unexpected imbalances between demand and supply” [1]. Kelly et al. provide an empirical discussion of just energy transition in their study of energy poverty mitigation in Ireland [89]. In short, socio-technical transition frameworks continue to be critical to the wider scholarly inquiry on energy transition as the literature recognizes that profound energy shifts involve multi-dimensional struggles in incumbent social, technical, and technological systems [78]. However, given the complex magnitude and numerous challenges of such transitions, the interdisciplinary engagement at this point is far from sufficient.

5. Just transition as governance strategies

The concept of governance is considered to have created the groundwork for transition studies and continues to maintain its popularity within the field. Moss suggests that governance approaches to transition studies are useful because they can capture “the growing complexity of institutional structures, political processes, and social relations involved in the collective pursuit of public, common, or individual interests” [90]. In a similar vein, Huang and Liu define transition governance as a “multi-dimensional, multi-level, multi-actor, and multi-phase governing process to facilitate systemic transformations of socio-technical regimes towards sustainability” [87].

Although governance in the name of sustainable development attempts to steer society toward a more sustainable future, we cannot take for granted that sustainability-oriented policies will produce just and equal results [90]. The fundamental conditions, namely high emissions, poverty and inequality, pose crucial governance challenges [8], including simultaneously addressing the politics of transition and ensuring a socially just transition [23]. Swilling et al. state that “a just transition is only possible if the overall goal is human wellbeing within a sustainable world” [91]. Similarly, Hopwood, Mellor and O'Brien [92] argue that sustainable development is about linking human equity to the environment. The principle of equity should be explicitly expressed in relevant policies [40]. This view is also echoed by other scholars [38,61] who argue that a clear equity language is not only important for justice advocates in the stage of policy design, but could ensure that the principle of equity is implemented as intended [40]. For Muttitt and Kartha, the important policy principles of just transition are economic

efficiency, meeting development needs, and effort-sharing by a fully supportive government [93]. In particular, the authors call for wealthier regions or countries to share burdens of the poorer nations so that the low-carbon transitions can be moved forward in a more just manner [93].

Turning to the governance approach of just transition, scholars typically attribute a successful just transition to support from a broad coalition of actors and to engagement with local initiatives. Cha et al. suggest that a just transition could be achieved through four governance strategies: strong governmental support, dedicated funding streams, diverse and strong coalitions, and economic diversification [14]. Newell and Mulvaney suggest that such coalitions could be built through engagement with social movements and community actors to develop an alternative vision and bring together a coalition of actors to provide and finance ‘just energy’ [23]. Such an approach is essential to overcome the difficult political tradeoffs that characterize a just transition. This view is echoed by Winkler, who also stresses that coalition is a foundational element for a just transition movement, which refers to creating alliances that include a range of stakeholders, such as labor unions, social movements, non-state actors, and some firms and international organizations [8]. The author further suggests that just transition as a broad concept allows different actors to make their own interpretations, which is not an obstacle if all support the objectives of a just transition.

Thus, the conceptualization of just transition governance is primarily conducted in the context of democratic regimes, where the role of network and participatory governance is heavily emphasized. Healy and Barry suggest that just transition should be considered as a form of “energy democracy” [7]. A strong coalition consists of a wide range of actors to reduce friction between various stakeholders with different interests and to allow for open and democratic participation by narrowing exclusion and inequality and permitting all voices to be heard [9]. This does not mean that just transitions happen in democratic regimes automatically, but to some extent, democracies do have institutional advantages to achieving a just transition owing to their political system [9].

Modes of governance, however, differ significantly across different socio-political contexts. Although governments can choose any combination of governance approaches, each government's style of governance is rooted in the existing political system [87]. The key question, thus, is how just transition as a democratic-framed governance concept could be applied in a non-democratic context. To answer this question, a small but growing body of literature has begun the exploration of a just transition approach in authoritarian regimes. Unlike the energy transition processes in liberal democracies characterized by municipal voluntarism, participation, and gradualness, the authoritarian governance model adopts a top-down manner that stresses policy efficiency, while leaving little space for participatory decision-making and context-based policy adjustments [87]. Furthermore, tight control of the media, non-governmental organizations, and academia, means that these non-state actors have limited freedom to raise social justice concerns. As such, disadvantaged groups in authoritarian regimes face immense difficulties in gaining a role in formal decision-making processes and governance institutions under the non-participatory policymaking process. Consequently, the interests and voices of vulnerable groups are rarely considered, which may lead to social catastrophe [94]. For example, empirical studies conducted in China, a typical authoritarian regime, have revealed the tremendous social injustice created by the ongoing low-carbon transition [87,94–97]. Studies manifest that the rapid energy transition in China, driven by powerful political-administrative interventions, has resulted in serious socio-economic issues, including unemployment, a decline of local tax revenues, and social unrest [96,97]. Lo [97] points out that rather than creating benefits, energy transition in extractive areas leads to severe social injustice due to the lack of policies to alleviate negative impacts from the central government. Scholars also stress that social injustices in authoritarian regimes are rooted in the imbalance of structural power between state-

owned and private enterprises [94]. Whereas many privately-owned firms were forced to shut down by China's energy transition policies, state-owned enterprises survived owing to their political influence and connection [94,97]. Furthermore, unlike the workers in state-owned enterprises who had better access to government support, the majority of workers in private companies could not receive proper compensation either from the government or the company [94].

Scholars therefore suggest that it is urgent for authoritarian regimes such as China to incorporate just transition measures to achieve environmental goals without placing an unfair burden on disadvantaged regions, communities and individuals [96–99]. The suggestions tend to focus on the concept of restorative justice, rather than the more politically controversial concept of procedural justice. For instance, Zhang and Wang suggest that the central and local governments should introduce targeted policies and measures to assist the regions and industries that are most affected [98]. Xu proposes that compensation policies be established to mitigate negative impacts on traditional energy industries during an energy transition [99].

In sum, the literature of just transition as governance strategies reveals how (un)just transition is shaped by political context. The debate over governance measures with regard to democratic and authoritarian systems highlights an important question: how to resolve the tension between the urgency of climate action and the inclusion of actors in decision-making? Transitioning away from fossil fuels is urgently needed because of climate change, but it is also important to leave sufficient time for affected industries, communities, and individuals to rebuild industries, institutions, infrastructure, and human capital to tackle the profound social-technical changes associated with such transitions [93]. However, many just transition debates focus on the process and the need to include all voices, without a sense of how to square the need for inclusion and deliberation with the necessary speed and acceleration to prevent further injustice associated with climate change [100,101]. In this sense, the policy efficiency offered by authoritarianism could be appealing, although the implication for social justice is highly concerning. Finding a middle ground between procedural justice and urgency concerns could be a challenging task for theorists and practitioners of just transition.

6. Just transition as public perception

Studying public perception and attitudes towards low-carbon energy transition is important public support and legitimacy is vital for renewable energy projects and policies to be deployed and implemented smoothly [102]. In addition, social injustices arising from low-carbon transitions can be detected from public opinions, especially those who are directly affected [103]. This final theme in the literature defines just transition from the perspective of public perception, acceptance, and support of transitions to low-carbon energy production.

A significant body of literature focuses on the public perceptions toward renewable energy in general [104–106] or specific to a certain type of low-carbon energy such as wind [102,107], solar [108–110], and nuclear [111–115]. Given that renewable energy is typically framed as a measure for addressing environmental and climate matters, studies have examined the relationship between environmental beliefs and attitudes towards renewable energy [103]. For example, Motz applies a hybrid discrete choice model with latent variables on stated preference data to study the preferences of households with respect to energy sources for electricity generation. The findings indicate that a higher environmental concern is associated with a strong preference for a renewable-based energy mix [116]. In contrast, a study that analyzes how residents living next to wind energy facilities form their attitudes towards energy transitions found that environmental beliefs play a very weak role in influencing individual' views on renewable energy [103]. Rather, the distance of their residence from wind turbines is closely related to the level of support for renewable energy. The authors thus suggested that policymakers should pay more attention to locally relevant social values

rather than simply framing renewable energy as an environmental issue to gain wider public support on renewable energy development at the local level [103].

Transitioning away from fossil fuel production is one of the central challenges facing coal-dependent communities. Understanding how these communities respond to ongoing or impending low-carbon energy transitions is thus important, especially as social opposition would impact the scale and pace of energy transitions [103]. Thus, public perception, acceptance, and support of low-carbon transitions in communities that depend on fossil fuels have been closely examined by scholars [27,36,117–120]. For instance, Cha found that mine closures in the Powder River Basin region in the United States did not change opposition to energy transition or increase support for a just transition due to the longstanding importance of coal to the region [119]. Furthermore, an empirical study regarding the level of support for energy transition among citizens was conducted in Germany. The study revealed that the level of unemployment adversely affects respondents' attitudes towards energy transition. As such, it is necessary to provide fair compensation to the victims of energy transition as an important trust-building measure between citizens and government [121].

To simultaneously achieve a just transition and address the problem of unemployment in the coal industry, the development of renewable energy has become a key measure [36]. Studies have argued that it is important to facilitate acceptance of renewable energy by residents in fossil fuels producing regions. Hazboun conducted semi-structured interviews with community representatives in two energy production areas in the state of Utah. Her study found that residents generally held negative attitudes toward renewable energy developments within their communities because they felt that efforts to transition away from fossil fuels unjustly punished the industries and communities in which they operated [36]. Other scholars have compared public perceptions of different energy sources rather than evaluating public perceptions or acceptance of a single source of energy. For example, Crowe and Li compared public perceptions and acceptance of coal, solar energy, and energy transition policies in Illinois, Texas, and Vermont. They found that a community's sense of place, rooted in place attachment, landscape memory, and land use history should be carefully considered in such studies [120]. Scholars have also acknowledged that adaptability in the face of economic vitality is a crucial factor in individuals' and communities' attitudes toward energy transition [117]. Procedural fairness is also important in enhancing community involvement, which plays a significant role in reducing tensions and facilitating more effective deliberation mechanisms that may gain support from the public [122].

In summary, this final theme in the literature is predominantly concerned with stakeholders' attitudes toward energy transition. These studies generally examine changes in stakeholders' attitude toward energy transition over time, or under specific conditions, explore how fossil fuel industry stakeholders adapt to energy transition efforts, and compare public and stakeholder perceptions across different regions and industries. This body of literature discovered that individuals' and communities' attitudes towards energy transitions could be influenced by a rich variety of factors such as the level of unemployment, environmental sensitivity, and an ingrained connection to the fossil fuel industry [57,116,121]. However, the role of public attitudes in clean energy transitions is unclear and worthy of more in-depth investigation.

7. Discussion

Several interesting insights can be garnered from analyzing the different trends in just transition literature. First, there are differences in understanding the causes of injustice. While there is a shared attempt to identify vulnerable groups in low-carbon transitions, different trends in the literature exhibit different understandings of who should be considered vulnerable. The labor-oriented concept was criticized as it focuses too much on the male-dominated fossil fuel workers, while other

affected groups were largely ignored and targeted. Consequently, just transition strategies mainly focus on unemployment compensation and skills training for obtaining new jobs [14,38]. Cha et al. criticize this approach by stating that a just transition must not just be compensatory but comprehensive [14]. In contrast, scholars adopting just transition as a holistic framework of justice see the groups most affected by low-carbon transition as low-income households, people with health concerns, and residents living in rural areas [9]. By adopting a more transformative notion of justice, these scholars seek to address the causes of injustice that are rooted in race, class, and gender. As such, the question of just transition for “whom” is still under debate. From a governance perspective, the problem of injustice is typically attributed to governance and political systems. This is particularly relevant to the governance of utility-scale energy systems because of the common phenomenon of lack of transparency and of public participation in the decision-making process [9]. However, it should be noted that injustices not only exist in large-scale, centralized energy projects, but decentralized forms of energy can also generate the unfair distribution of benefits and harms. It should also be noted that these different interpretations of causes are not mutually exclusive, but rather, are interrelated.

Second, and relatedly, we note the tension between different understandings of just transition. While the labor-focused concept of just transition has strong connections to environmental justice and climate justice [19,40,42], there are nevertheless tensions between these different concepts of just transition. For example, Kalt points out that labor movements and climate movements hold inherently opposite positions on jobs and the environment as climate movements seek to leave fossil fuels in the ground, whereas labor unions oppose phasing out fossil fuels in order to protect high-paying jobs for fossil fuel workers [39]. The inability to resolve the conflicts between environmental protection and maintaining fossil fuel-based jobs means that, eventually, the concept of just transition turns to a “fancy name for a funeral” and becomes allied to fossil fuel capitalism [38]. In addition, the tension is also mirrored within the diversity of labor unions or even the contradictory meanings of just transition given the different levels of utilization of strategy for just transition by international labor unions within a wide range of political contexts around the world [38].

Third, in general, sophisticated empirical studies are lacking as scholars seem more interested in creating various just transition analytical frameworks than in applying these frameworks to empirical investigations. The focus is often on what should happen or who should be included in the process, rather than on the efficacy and problems of just transition frameworks in action, or what was it about transitions to date that made them unjust. Further, little is written on the ways in which the just transition framework is co-opted or limited to prioritize the interests of certain strategies, interest groups, and industries. The literature would benefit from research that examines how competing ideas of just transition are handled in real-world settings, as well as how governance strategies evolve as just transitions gather momentum.

Fourth, future studies should pay more attention to the power dynamics in just transition. Several scholars assert that multiple actors from government, international institutions, business, and civil society play an essential role in mobilizing low-carbon transition strategies. However, critical examinations of the power relations among these stakeholders are largely absent from the just transition literature. Future studies should examine how just transitions are discursively constructed to form coalitions across a variety of actors, should determine how actors are strategically mobilized, and should investigate how power relations change (or are maintained) between incumbent and emergent actors over time. The underlying principle of just transition is equity, which aims at both empowering the weak and weakening the powerful at the most general level of social justice. However, the concept of just transition could be co-opted by powerful actors to prioritize certain strategies, social groups, and industries. The extent to which just transitions can achieve a radical transformation in power relations is unclear.

Fifth, given that political contexts play an important role in shaping the power dynamic and governance of just transition, further research is needed to explore the relationship between different political systems and different types and outcomes of just energy transition. It would be beneficial, for instance, to take a comparative approach that uncovers a range of similarities and differences between energy transitions taking place in democratic and authoritarian regimes.

Sixth, future studies should expand their geographic scope to better understand how transitions unfold across different geographical contexts and why the success and implementation of justice-oriented transition policies are context-dependent. Just transition research focuses on the labor–environment nexus. However, the suggested harmonious relation between the two is often based on a well-developed “green economy.” Just transition studies mainly focus on a few industrialized countries such as the United States, Germany, United Kingdom, and Australia [27,36,117–120,123]. It would be useful for the literature to include an examination of the concept and application of just transitions in developing countries, as addressing justice issues in these countries poses an even greater challenge, particularly because regions or countries dominated by mineral and fossil fuel resources face difficulties in achieving socio-economic development and ecological goals simultaneously [124]. This has been proven in the few studies that focus on just transition in developing countries; for example, Swilling et al. conclude that a just transition in South Africa is currently unlikely because the country’s political economy has been dominated by mineral and fossil fuel extraction and the political elites have committed serious rent-seeking and corrupt practices [91]. The key challenge for scholars is thus to incorporate the spatial dimension into an understanding of just transition, which could stimulate theoretical innovation and refinement. Moreover, just transition practices in different parts of the world may shape how a just transition should be pursued at a global level [23]. As noted by Bridge, Özkaynak and Turhan [125], the world is experiencing transformational growth in energy infrastructure. However, at the global level, low-carbon energy transitions are a spatially uneven process affected by geopolitics and could result in the uneven displacement of injustices onto poorer countries [37,126]. Addressing these issues is a key challenge for future work on just transitions.

8. Conclusion

To consolidate knowledge, advance the debate over just transition and our understanding of it, this review focuses on how this term with a distinctive historical backdrop has evolved and been adopted in academic circles. Five themes have been identified. First, scholars focusing on the historical roots of just transition understand the concept primarily as a labor-oriented construct that links energy transition with economic transition. Second, some scholars, recognizing the inherit link between just transition and other justice concepts, particularly environmental justice, climate justice, and energy justice, envision just transition as an integrated framework of justice. Third, there are some attempts to develop just transition as a framework for cooperating justice principles into socio-technical transition studies, thereby bringing attention to the material dimensions of just transition. Fourth, just transition can be understood as a governance phenomenon, and scholars taking this approach focus on how just transition unfolds under different governance regimes. Fifth, inspired by the idea that justices or injustices are rooted in “the meaning of life” [53], scholars in this theme explore the public perceptions and attitudes surrounding the justice of low-carbon transitions. The review shows that, given the proliferation of just transition research in recent years, the term has become extended, multifaceted, and to some degree problematically polysemic, which leaves room for confusion in interpretation. It is thus important to conceptually distinguish the different strands of just transition, as well as to recognize the tension between the different strands.

This paper contributes to the literature and practice in several ways. First, it serves as a toolkit for researchers and students in need of

comprehensive insight into the concept of just transition. Second, given that the concept is still very much in a development phase, this is an opportune time to further the debate on just transition and to expand its scope and relevance as a research field. This paper aims to advance a research agenda for just transition that can help to illuminate pathways forward for a more just future. Third, justice transition strategies have been practiced in many different regions and countries around the world and these experiences serve as lessons for policymakers. The articulation of just transition principles and strategies could have an impact by making policymakers, who often have too little regard for the long-term socioeconomic consequences, understand the core values underpinning the concept [63]. Fourth, this paper may prove to be useful to practitioners and activists interested in social justice, including labor unions, environmentalists, and climate justice activists. The review shows that the synergies among these different groups, as well as their common civil rights roots, indicate that the concept of just transition may create opportunities to unite these groups to overturn injustices associated with low-carbon energy transition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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